Udyam Express — solution architecture

prototype components (solid) · production integrations (mocked with real API shapes)

CHANNELS

Owner web app (PWA)
Phone-first chat + voice (9 Indian locales), quote, KYC, tracker

WhatsApp / IVR
Same advisor API behind WhatsApp Business API (production)

Partner / field tablet
Assisted mode: same flow, commission credited to the partner

Back office
Hand-offs, live orders, win-back queue, funnel

UDYAM EXPRESS PLATFORM (FASTAPI)

AI talks · rules decide

1 · Advisor
Rule-based NLU (EN / हिन्दी / Hinglish) + optional Gemini extraction & phrasing. Hand-off rules: tier-3, >10 sites, custom needs.

nlu.py · advisor.py · llm.py

2 · Rules engine & OneBill
Catalogue eligibility (tier, staff, sites, GST), dependencies, 3 tiers, midpoint pricing, GST shown separately, 24-mo setup waiver.

core.py · recommender.py

3 · Smart KYC
Consent ledger (hash-chained) → OCR on own server (deskew, blur gate, GSTIN look-alike fix + checksum) → registry & consistency checks → 4 verdicts.

kyc.py

4 · Orders & tracker
6 stages, ETA per product from install days, SLA risk flag, WhatsApp updates.

backoffice.py

Propensity & win-back
Logistic regression (AUC 0.72, June hold-out) × drop-reason recoverability → next best action.

Privacy layer
PII redaction before storage/LLM, masked IDs in UI & logs, 30-day document purge, withdraw & erase.

Flow: chat → profile → rules engine → quote (toggles re-priced server-side) → consent → KYC verdict → e-sign & UPI AutoPay → order → tracker. Unclosed chats → lead with summary → partner.

DATA & INTEGRATIONS

Catalogue (25 products)
Single source of truth for products, prices, rules

GSTIN & Udyam registries
Mock lookups from CSV → GSTN / Udyam APIs in production

5,000 historic leads
Trains the propensity model; powers the funnel & impact replay

Gemini (optional)
Gets only redacted text; never documents, never decides price

DigiLocker · Aadhaar e-Sign · UPI AutoPay · Field-ops
Production integrations (mocked in prototype)

Stack: Python 3.11 · FastAPI · pandas · scikit-learn · OpenCV + Tesseract · vanilla JS PWA · Docker

Runs offline at ₹0 — Gemini key optional

27 tests · 0 rule violations on 5,000 profiles · 12/12 KYC